



Philippine Christian University



Sampaloc I, Dasmariñas, Cavite

<https://dasma.pcu.edu.ph>

Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**

3. Select **Bridged** mode

4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SSH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SSH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
        valid_lft 179068sec preferred_lft 83386sec
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtmpdr proto kernel_r
        valid_lft 179068sec preferred_lft 92668sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:                0.0
Usage of /:                  5.1% of 48.91GB
Memory usage:               5%
Swap usage:                 0%
Processes:                  139
Users logged in:            1
IPv4 address for enp0s3:    192.168.1.14
IPv6 address for enp0s3:    2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of `ip a` or `hostname -I`)
- SSH service status (`sudo systemctl status ssh`)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

Name: Geisler A. Borquillo_____

Student ID: 202202471_____

Course/Year/Section: BSIT 3B _____
Date Submitted: 02/09/2026 _____
Instructor: Johneros P. Puyo _____

1 OBJECTIVES (5 POINTS)

Check all that apply:

- ☐ Create and configure a Virtual Machine with Ubuntu Server
- ☐ Configure Bridged Network Adapter for network access
- ☐ Determine and verify the server's IP address
- ☐ Install OpenSSH Server and enable the service
- ☐ Establish successful SSH connection from host machine
- ☐ Execute verification commands to confirm remote access

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Ubuntu
Virtualization Software (VirtualBox/VMware)	Oracle VM VirtualBox
Ubuntu Server Version	26.04
VM Name	geislerserver
RAM Allocated	6.7GB
Storage Allocated	105GB
Network Mode	Bridged Adapter
SSH Client Used	Secure shell client

3 VM CONFIGURATION

Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)
Memory	___6703___ MB
Hard Disk	___105.00___ GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION

Commands Executed: `sudo apt update`
`sudo systemctl enable ssh`
`sudo systemctl start ssh`
`ip a`

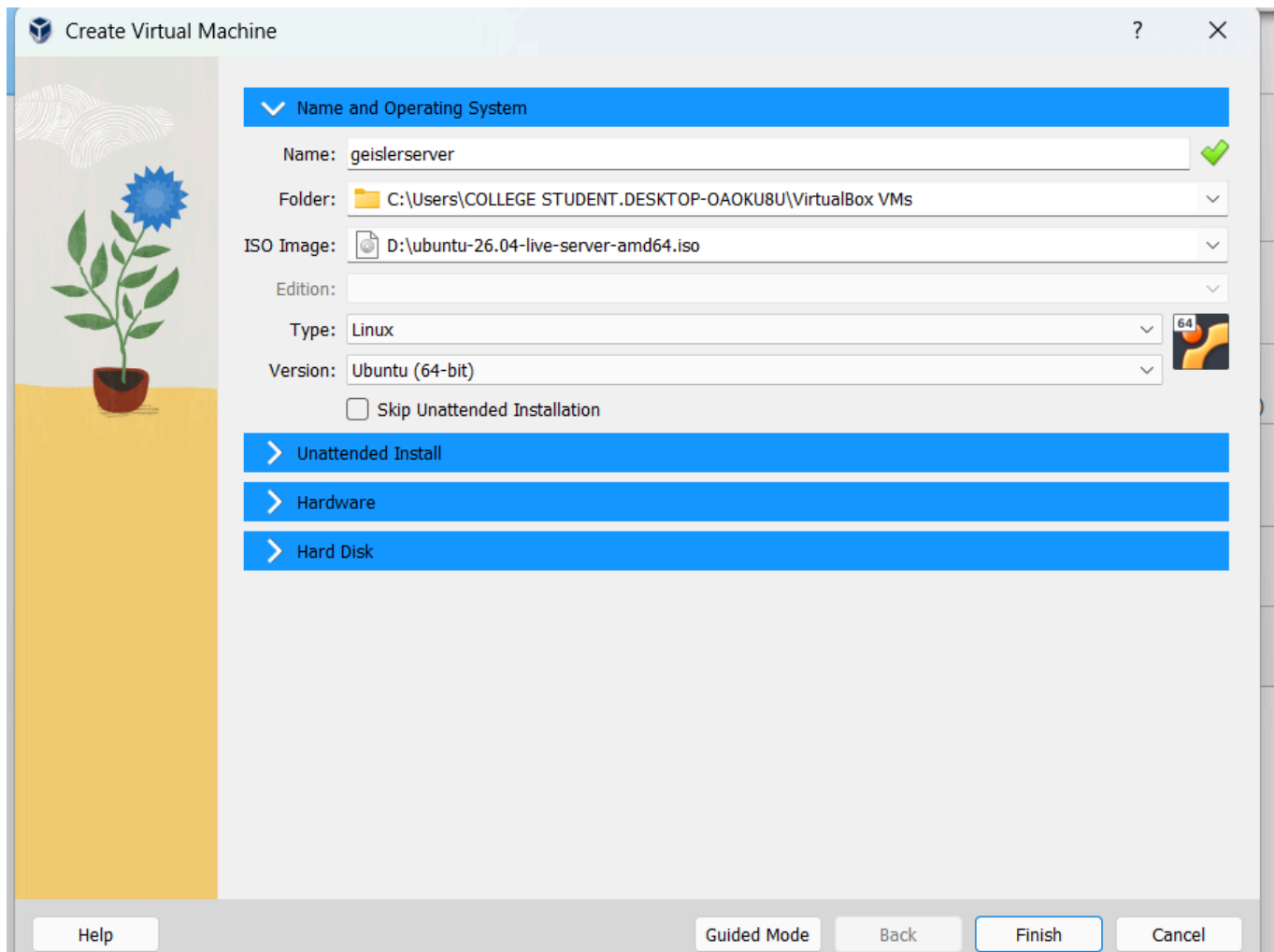
Server IP Address: 192.168.137.132_____

Network Interface: Bridged Network_____

6. REMOTE CONNECTION

Connection Details:

Parameter	Value
Server IP	192.168.137.132
Username	sylph12
SSH Port	ssh sylph12@192.168.137.132



Create Virtual Machine

▼ Name and Operating System

Name: ✓

Folder: ▼

ISO Image: ▼

Edition: ▼

Type: ▼

Version: ▼

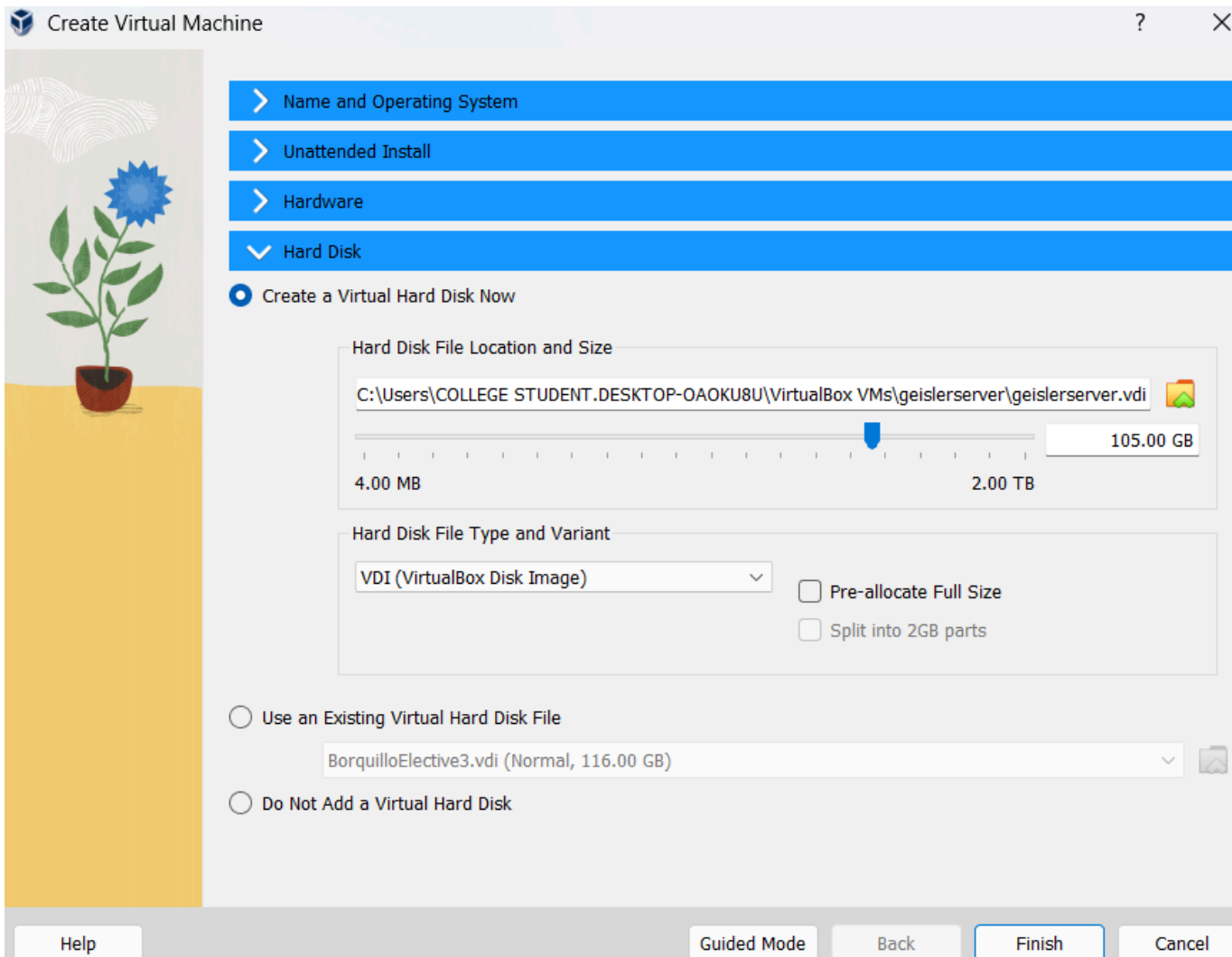
☐ Skip Unattended Installation

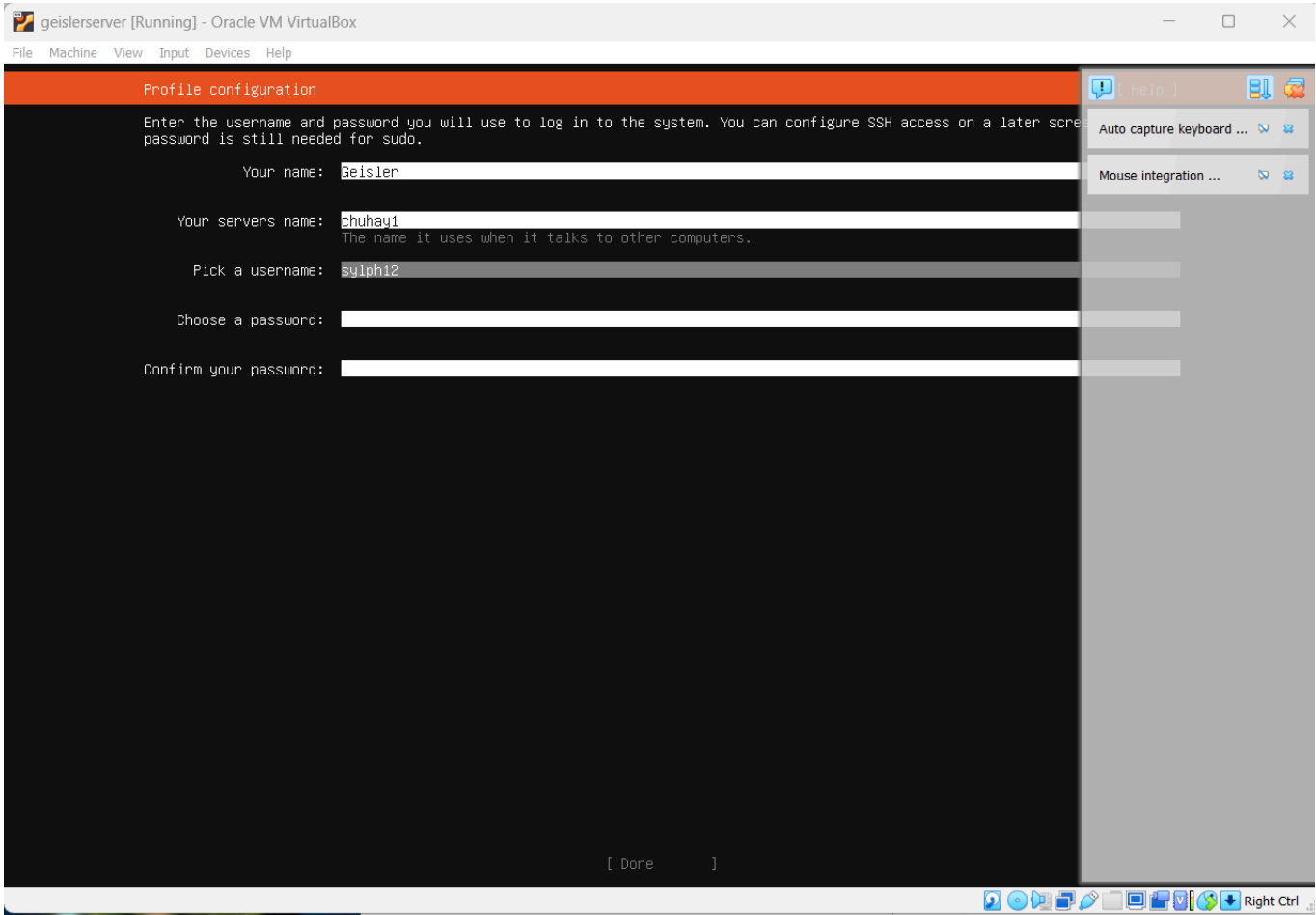
> Unattended Install

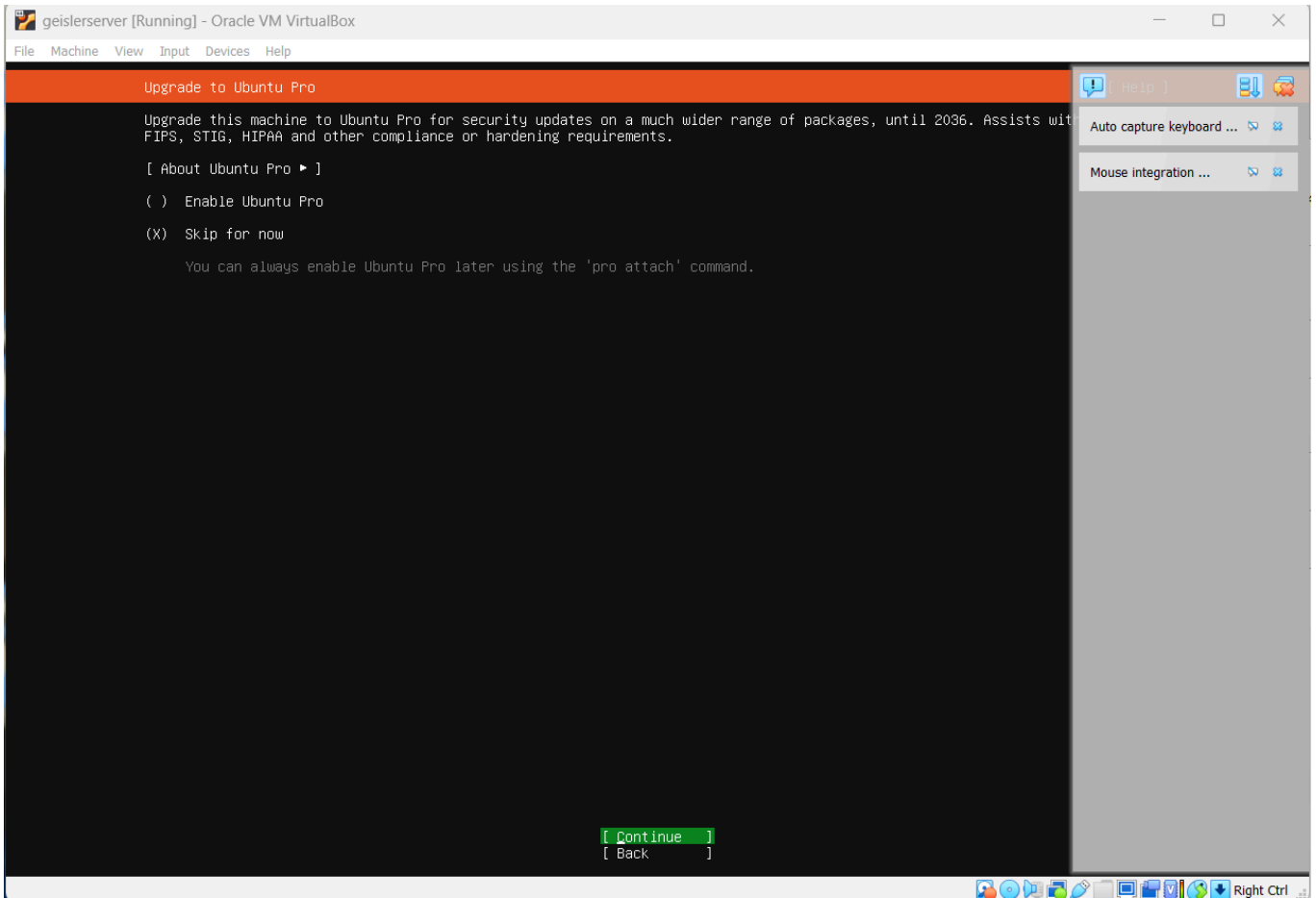
> Hardware

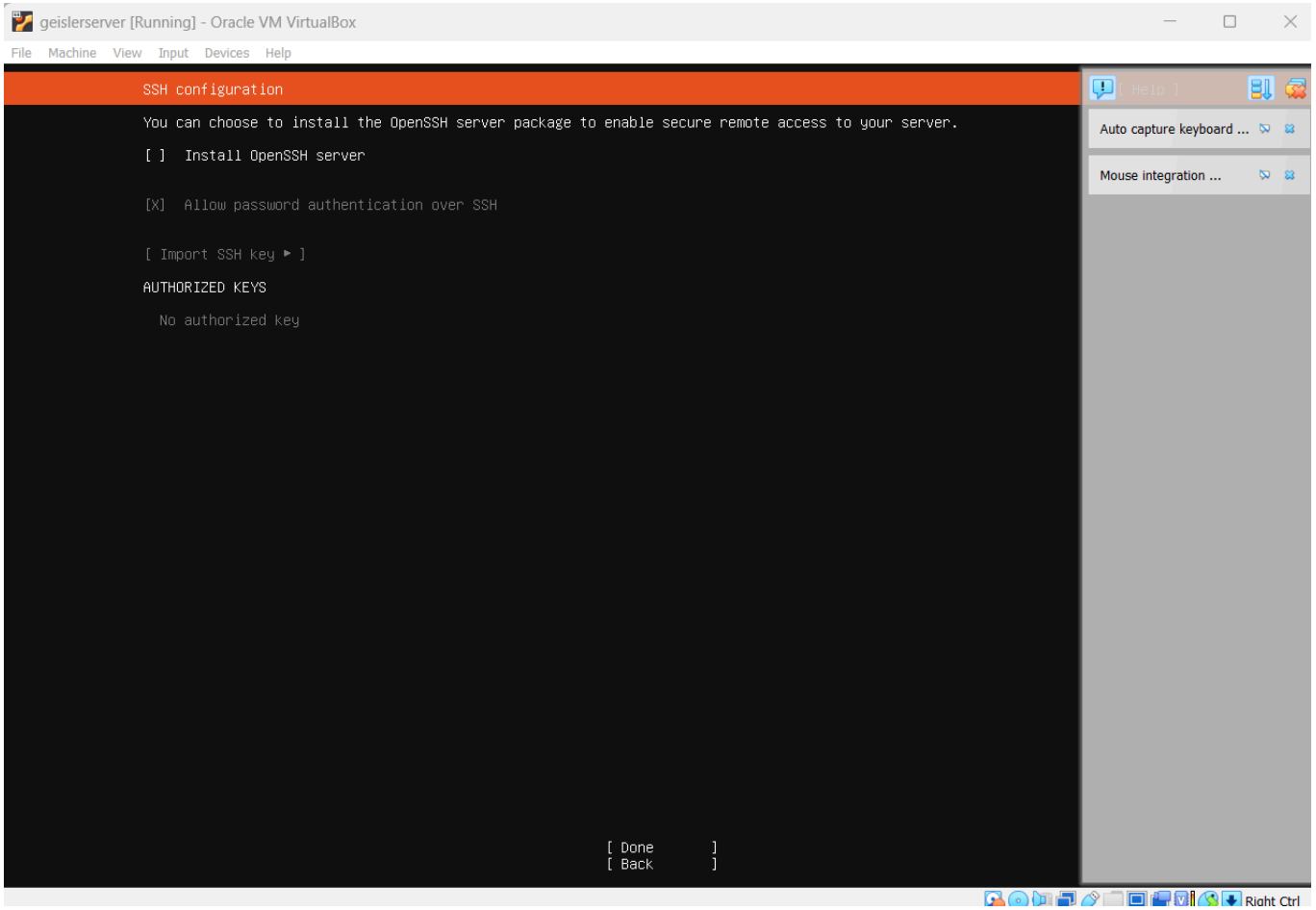
> Hard Disk

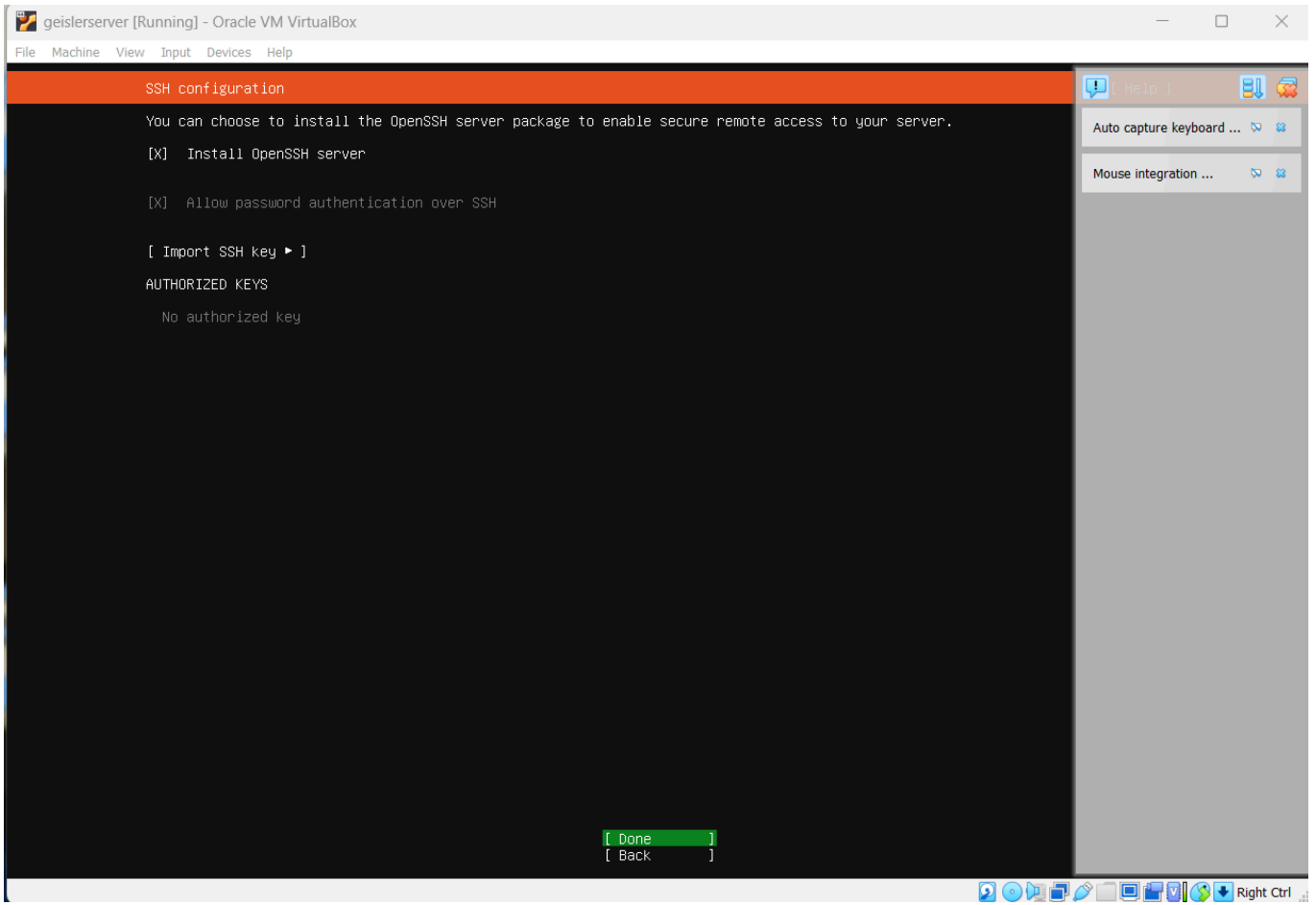
Help Guided Mode Back Finish Cancel

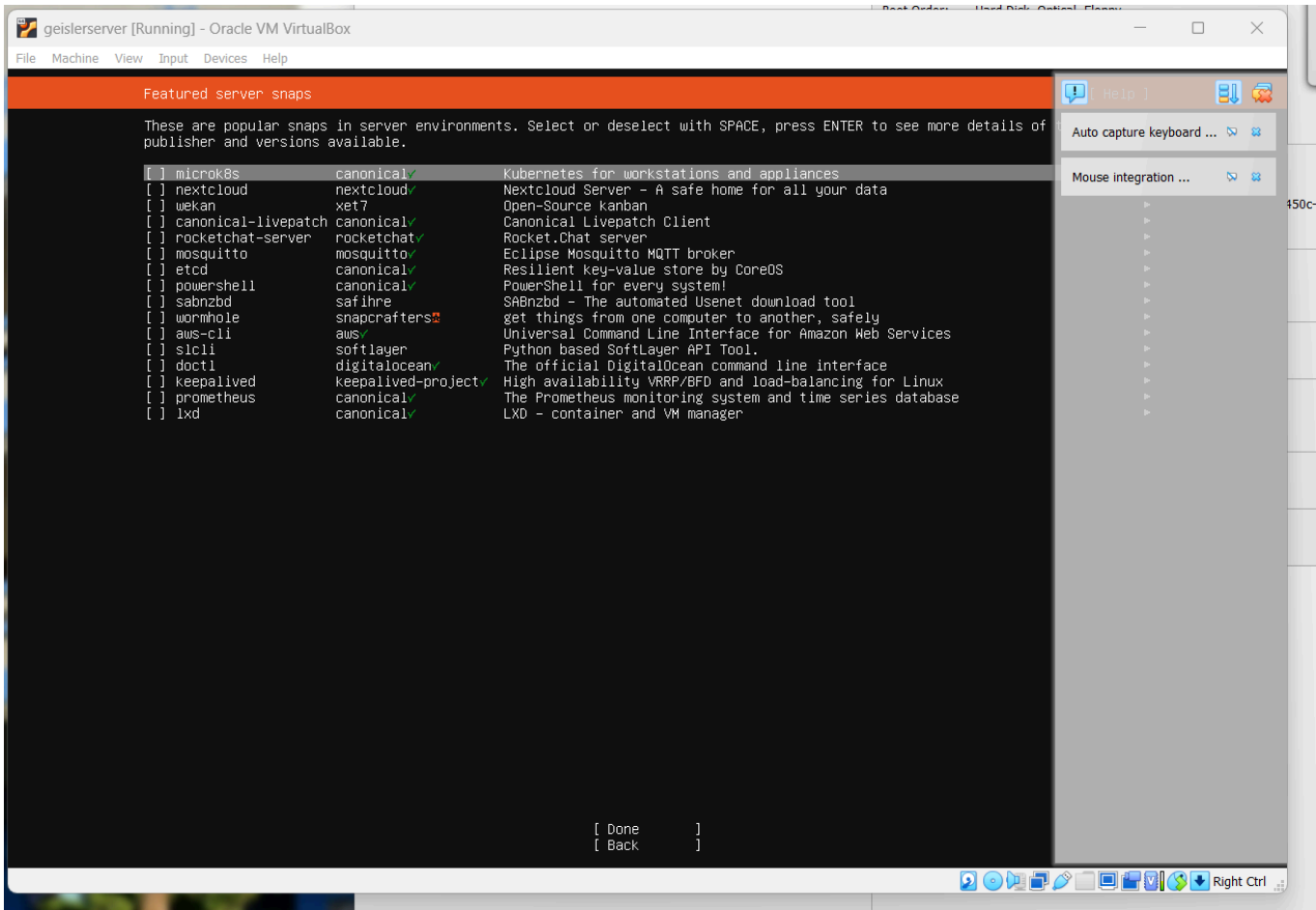


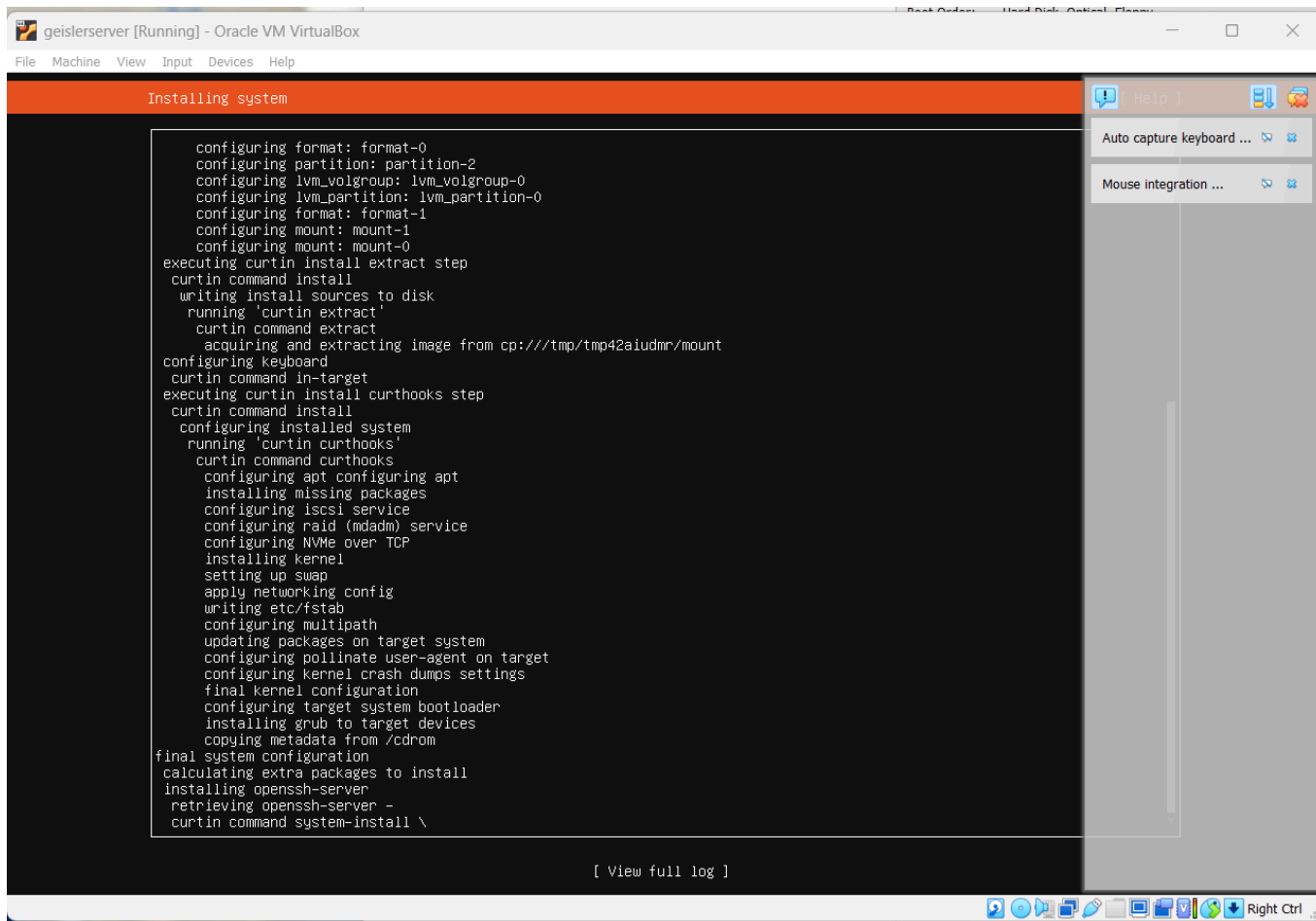


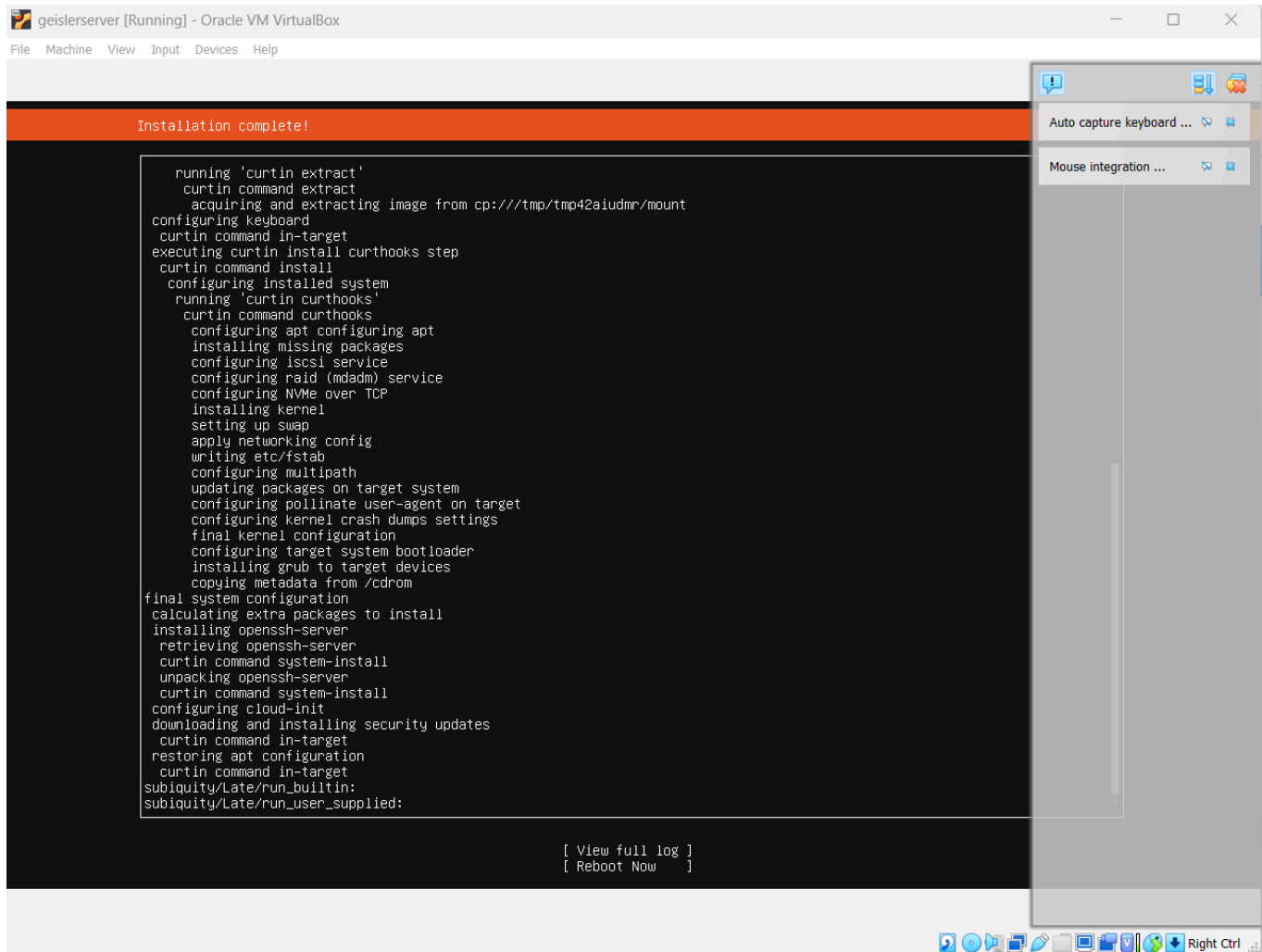


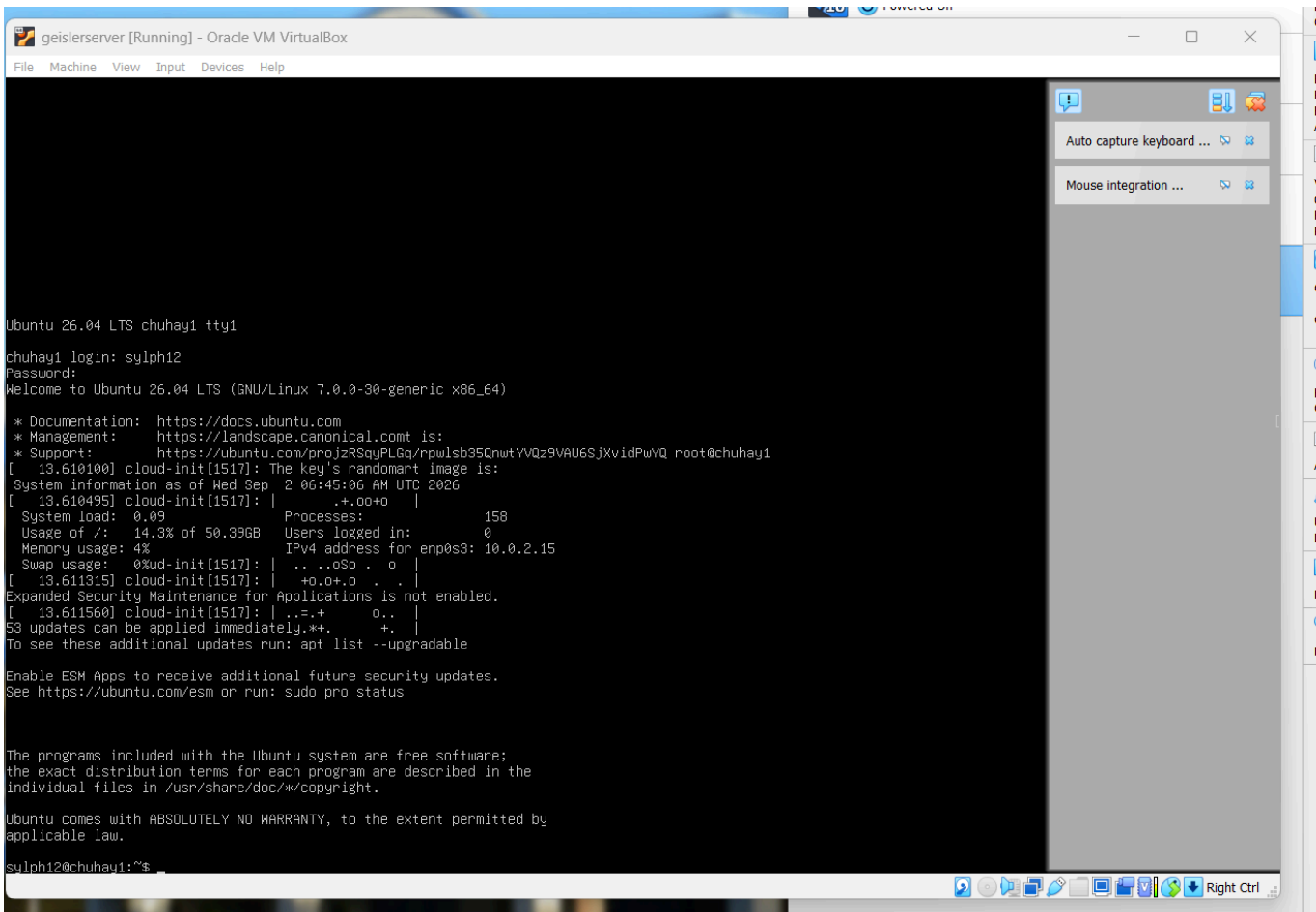












geislserver [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

```
Ubuntu 26.04 LTS chuhay1 tty1
chuhay1 login: sylph12
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/projzRSqyPL6q/rpwlsb35QnwtYVQz3VAU6SjXvidPwVQ root@chuhay1
[ 13.610100] cloud-init[1517]: The key's randomart image is:
System information as of Wed Sep 2 06:45:06 AM UTC 2026
[ 13.610495] cloud-init[1517]: | .+.oo+o |
System load: 0.09 Processes: 158
Usage of /: 14.3% of 50.39GB Users logged in: 0
Memory usage: 4% IPv4 address for enp0s3: 10.0.2.15
Swap usage: 0%ud-init[1517]: | ..ooSo . o |
[ 13.611315] cloud-init[1517]: | +o.o+.o . . |
Expanded Security Maintenance for Applications is not enabled.
[ 13.611560] cloud-init[1517]: | ..=+. o.. |
53 updates can be applied immediately.*+. +. |
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

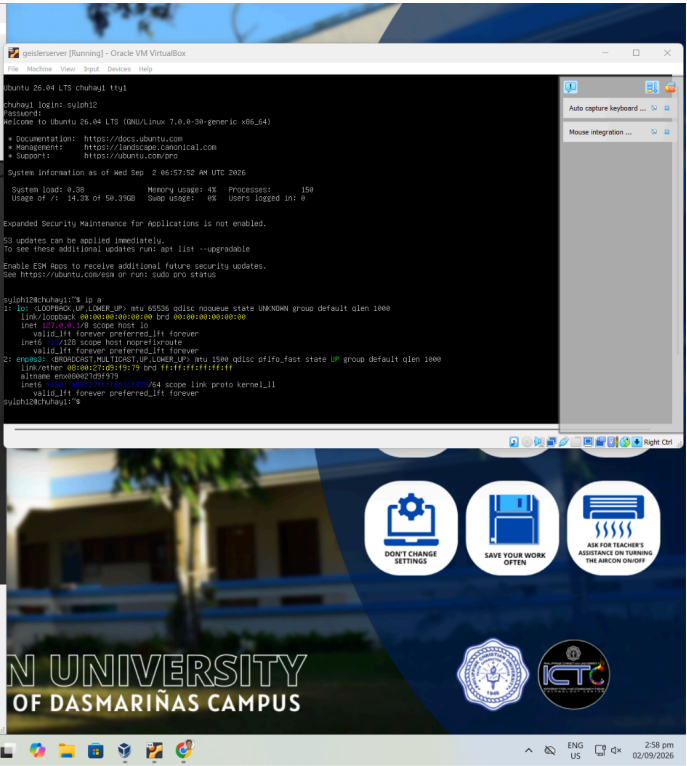
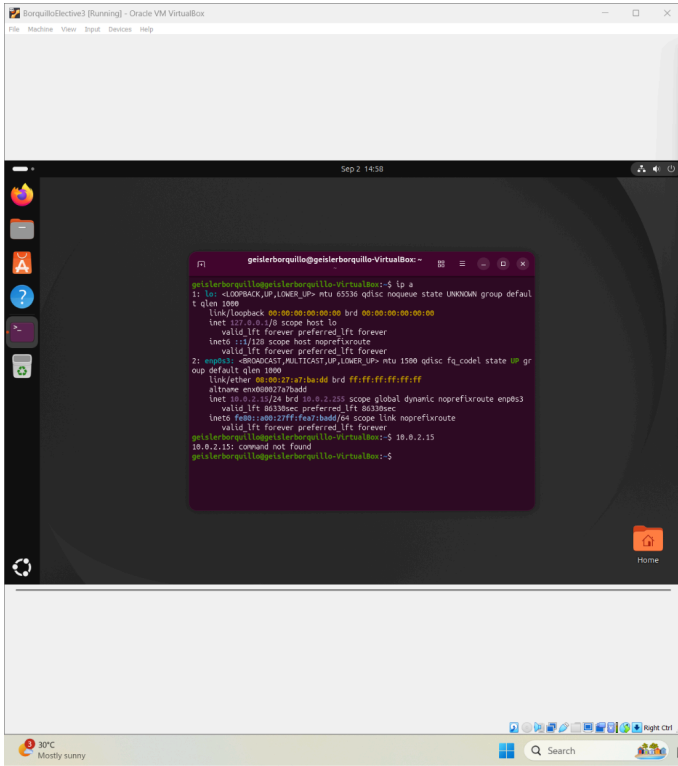
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

sylph12@chuhay1:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:d9:f9:79 brd ff:ff:ff:ff:ff:ff
    altname enx080027d9f979
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86334sec preferred_lft 86334sec
    inet6 fe80::a00:27ff:fe99:f979/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
sylph12@chuhay1:~$
```

Auto capture keyboard ...

Mouse integration ...

Right Ctrl



geiserserver [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

```
* Documentation: https://docs.ubuntu.com
* Management:   https://landscape.canonical.com
* Support:      https://ubuntu.com/pro

System information as of Wed Sep 2 07:53:30 AM UTC 2026

System load:  0.01      Processes:      158
Usage of /:   14.5% of 50.39GB  Users logged in: 0
Memory usage: 4%      IPv4 address for enp0s3: 192.168.137.132
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

sylph12@chuhay1:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
55 packages can be upgraded. Run 'apt list --upgradable' to see them.
sylph12@chuhay1:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
sylph12@chuhay1:~$ sudo systemctl start ssh
sylph12@chuhay1:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:d9:f9:79 brd ff:ff:ff:ff:ff:ff
    altname enx080027d9f979
    inet 192.168.137.132/24 metric 100 brd 192.168.137.255 scope global dynamic enp0s3
        valid_lft 604718sec preferred_lft 604718sec
    inet6 fe80::a00:27ff:fed9:f979/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
sylph12@chuhay1:~$
a      apt      enable      exiexit    exit        ip          ssh         start       sudo        systemctl  update
sylph12@chuhay1:~$
```

Auto capture keyboard ...

Mouse integration ...

Right Ctrl

Sep 2 15:58

```
sylph12@chuhay1: ~ — ssh sylph12@192.168.137.132
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a7:ba:dd brd ff:ff:ff:ff:ff:ff
    altname enx080027a7badd
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83983sec preferred_lft 83983sec
    inet6 fe80::a00:27ff:fea7:badd/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
geislerborquillo@geislerborquillo-VirtualBox: ~$ ssh sylph12@192.168.137.132
The authenticity of host '192.168.137.132 (192.168.137.132)' can't be established.
ED25519 key fingerprint is: SHA256:HjzRSqyPLGq/rpwlsb35QnwtYVQz9VAU6SjXvidPwYQ
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.137.132' (ED25519) to the list of known hosts.
sylph12@192.168.137.132's password:
Permission denied. please try again.
```



Sep 2 15:59



```
sylph12@chuhay1: ~ — ssh sylph12@192.168.137.132
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Wed Sep  2 07:58:39 AM UTC 2026

System load:  0.11           Processes:            146
Usage of /:    14.6% of 50.39GB Users logged in:         0
Memory usage:  4%            IPv4 address for enp0s3: 192.168.137.132
Swap usage:    0%

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

sylph12@chuhay1:~$
```

